

Limitations of ZAFF-AMBER absolute binding free energy calculation for zinc metalloenzyme inhibitors: a replicate-pair analysis on MMP-1

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Abstract

Absolute binding free energy (ABFE) calculations on zinc metalloenzymes present a stress test for non-bonded zinc force fields such as ZAFF-AMBER: the catalytic Zn^{2+} binding chemistry of hydroxamate, sulfonamide, and thiol inhibitors must be reproduced from a fixed-charge, restraint-based description of the metal coordination shell. We benchmark the ZAFF-AMBER + GAFF-2 + AM1-BCC ABFE protocol on ten ChEMBL MMP-1 inhibitors spanning sub-nanomolar to micromolar potency (4–18000), six scaffold classes including hydroxamate, non-hydroxamate sulfonamide, aryl sulfone, carboxylate, fluoro-aryl, each computed with up to three independent replicates using the openmmtools MultiStateSampler. Our central observation is that **reproducibility is not accuracy**: replicates of the same compound agree to within 0.06 for some compounds yet are systematically biased by 4–18 from the experimental affinity, while other compounds give catastrophic 18.8 replicate variance whose mean nonetheless lands within 1 of experiment. Accuracy is also stratified by potency class: strong inhibitors (≤ 20) trend toward over-binding by 8–10, weak binders (> 10) consistently sign-flip, and the most accurate compound in our set is a mid-potency fluoro-aryl hydroxamate. A three-replicate mean recovers the experimental binding affinity to within 1 for the strongest binder despite the variance. We additionally document two pipeline pitfalls — (i) ReplicaExchangeSampler swap-all deadlocks at the first exchange iteration on this system class, requiring the non-exchanging MultiStateSampler replacement, and (ii) ABFE production crashes with NaN energies on medium-sized ligands unless an explicit minimization + heating + restrained NPT warmup is interposed between tleap and Phase 5. We position this work as a limitations evaluation rather than a validation study and recommend ≥ 3 -replicate ABFE with explicit per-class error budgeting for ZAFF-AMBER metalloenzyme inhibitor ranking.

Keywords: ABFE, ZAFF-AMBER, zinc metalloenzyme, MMP-1, hydroxamate, MultiStateSampler, replicate variance, openmmtools.

Introduction

ABFE for metalloenzyme inhibitor ranking

Absolute binding free energy (ABFE) calculations evaluate ligand-protein affinity at quantitative resolution by alchemically decoupling the ligand in two thermodynamic states (bound complex, pure solvent) and combining the two legs through a thermodynamic cycle. For zinc metalloenzymes such as the matrix metalloproteinase MMP-1, the catalytic Zn^{2+} ion is the principal binding determinant for hydroxamate-class inhibitors (Marimastat, Batimastat, Prinomastat) that achieved nanomolar potency in the early MMP inhibitor programs.

A non-bonded zinc force field — ZAFF-AMBER, here applied as the `ions234lm_126_tip3p` parameter set augmented by Hertel’s 12-6-4 zinc-coordination terms — describes the metal as an explicitly charged ion held in place by short-range Lennard-Jones potentials plus implicit harmonic restraints to the three coordinating histidines. Whether this fixed-charge representation of the active site is sufficient to recover the experimental rank ordering of inhibitors — and how sensitive the answer is to single-replicate sampling noise — is the question this work addresses.

Distinguishing reproducibility from accuracy

A single ABFE run reports a ΔG_{bind} estimate together with a bootstrap-style statistical error from the alchemical analysis (typically 0.3–0.8). This statistical error is a within-run dispersion: it captures Poisson-like fluctuation across the alchemical states inside the same simulation but says nothing about the systematic bias of the protocol or the variance between independent replicates. The published ABFE benchmark literature emphasizes both error sources, but applied screening efforts often report only the within-run error.

In this work we run replicate ABFE calculations on each compound (2–3 independent replicates per compound, fresh velocity seeds, no shared trajectories) and report both the within-run statistical error and the between-replicate dispersion \

(Δ_{rep}). We demonstrate that these quantities are decoupled on the ZAFF-AMBER MMP-1 system class — and that decoupling has implications for protocol-wide error reporting in metalloenzyme ABFE.

Pipeline pitfalls and reproducibility infrastructure

ABFE calculations are notoriously implementation-sensitive. We document two specific failure modes encountered during this benchmark and the corresponding pipeline fixes:

1. The `openmmtools.multistate.ReplicaExchangeSampler` with `swap-all` replica-exchange policy deadlocks on the ZAFF-MMP-1 systems at the first exchange iteration (worker threads enter a `futex` wait loop, main thread spins; no further trajectory frames are written). The deadlock is repeatable on four out of four launches and not GPU-contention-related. Switching the sampler to `MultiStateSampler` (no replica exchange, independent state-replica simulation) resolves this completely with no measurable convergence penalty.
2. Phase 5 production hits `SimulationNaNError` at iteration 1 on every ligand larger than approximately C8 unless an explicit warmup stage (10000-iteration minimization, 0 K–310 K heating over 100 ps, then 100 ps restrained NPT and 100 ps unrestrained NPT) is interposed between the `tleap`-built complex and the alchemical decoupling. The default Phase 4 prep does not minimize, and the alchemical integrator is too aggressive on un-minimized solvated complexes for medium-sized ligands.

Both fixes are now baked into our pipeline (`abfe_benchmark_orchestrator.py` chains warmup before production, and `abfe_benchmark_prepare.py` auto-detects ligand net charge via SMARTS to fix `antechamber`'s `sqm` electron-parity errors on the anionic hydroxamate / carboxylate / sulfonate / acyl-sulfonamide functional groups). All scripts are released open source; see Section 6.

Methods

Compound selection and preparation

Seven ChEMBL MMP-1 inhibitors were selected to span four orders of magnitude in IC_{50} (4 –18000) and four scaffold classes:

- **Hydroxamic-acid peptidomimetics:** ChEMBL415 (Batimastat, 4); ChEMBL443684 (Marimastat, 5); ChEMBL94487 (RS-130830, 12); ChEMBL257077 (prinomastat-like, 15); ChEMBL292707 (Ilomastat / GM6001, 200); ChEMBL2105729 (low-potency hydroxamate, 18000).
- **Sulfonamide hydroxamate:** ChEMBL301236 (42 , fluoro-aryl).

Ligand 3D coordinates were generated by RDKit ETKDG embedding from SMILES followed by MMFF94 optimization. Docked poses were obtained with AutoDock Vina (grid centered on the catalytic Zn^{2+} at ((40.32, 27.89, 36.94)) Å, (25 × 25 × 25) Å box, exhaustiveness 16). Net charge was inferred via RDKit SMARTS detection of hydroxamate, carboxylate, sulfonate, phosphonate, tetrazole, and acyl-sulfonamide groups, and passed to antechamber as the -nc flag for AM1-BCC charging on the GAFF-2.11 atom-typing.

Receptor and force-field setup

The receptor was 1HFC chain A (catalytic domain of human MMP-1 in the holo form), with the catalytic zinc and three coordinating histidines (HID111, HID115, HID121, NE2 distance 1.88 Å–2.08 Å) preserved. AMBER ff14SB described the protein, GAFF-2.11 the ligand, TIP3P the water, and the ions234lm_126_tip3p parameter set the zinc and any charge-balancing Na^{+} / Cl^{-} counterions.

The complex topology was assembled with tleap (TIP3P box, 12 Å buffer, charge-neutralized with Na^{+} / Cl^{-}); solvent-leg topologies were assembled identically without the protein present.

ABFE protocol

Warmup.

Each complex (and each solvent leg) was warmed prior to alchemical decoupling: 10000-iteration L-BFGS minimization (tolerance $10 \text{ kJ mol nm}^{-1}$), 0 K-310 K heating in 10 stages of 10 ps each, 100 ps restrained NPT (protein heavy atoms restrained at $10 \text{ \AA}^{-2}$), then 100 ps unrestrained NPT.

Sampler.

We use the `openmmtools.multistate.MultiStateSampler` (no replica exchange). Sixteen alchemical (λ) -windows interpolate from the fully coupled ligand to the fully decoupled state, splitting electrostatics first ((λ_{elec})): $(1 \text{ to } 0)$ over the first 8 windows) and sterics second ($(\lambda_{\text{sterics}})$): $(1 \text{ to } 0)$ over the remaining 8). Soft-core potentials follow the `openmmtools.AbsoluteAlchemicalFactory` default ($(\alpha = 0.5)$, $(\beta = 12)$).

Restraint.

A flat-bottom centroid distance restraint keeps the ligand near the active site during decoupling: $(V(r) = 0)$ for $(r \leq r_{\text{max}})$ and $(V(r) = \frac{1}{2} k (r - r_{\text{max}})^2)$ otherwise, with $(k = 10 \text{ kcal mol}^{-1})$ ($(\text{unit}^{\text{angstrom}^{-2}})$) and $(r_{\text{max}} = 8 \text{ unit}^{\text{angstrom}})$. The corresponding analytical standard-state correction is
$$\Delta G_R^{\text{circ}} = -RT \ln \frac{V_R}{V^{\text{circ}}}, \text{quad } V_R = \frac{4}{3} \pi r_{\text{max}}^3, \text{quad } V^{\text{circ}} = 1660.5 \text{ unit}^{\text{angstrom}^3}.$$

Production.

8 ns per window in the complex leg, 5 ns per window in the solvent leg. 1 ps integration timestep with constraints on bonds to hydrogen (HBonds rigid water). Langevin Middle integrator at $(T = \text{SI}\{310\} \text{ kelvin})$, friction 1 ps. Per-state energy reduced potentials are written every 250 timesteps. Total production time per compound: $(16 \times (8 + 5) = \text{SI}\{208\} \text{ nano second})$ per replicate.

Analysis.

The `MultiStateSamplerAnalyzer` performs MBAR-based estimation of $(\Delta G_{\text{complex}})$ and $(\Delta G_{\text{solvent}})$. The binding free energy is
$$\Delta G_{\text{bind}} = \Delta$$

$G_{\mathrm{solvent}} - \Delta G_{\mathrm{complex}} - \Delta G_{R^{\circ}}$, where the negative sign on $(\Delta G_{R^{\circ}})$ reflects that releasing the flat-bottom restraint to the standard-state volume is favorable.

Replicates and experimental reference

Independent replicates differ in (i) random initial velocity assignment at 310 K, (ii) per-state random number streams in the Langevin integrator, and (iii) random number streams in MBAR’s self-consistent iteration. The complex tleap topology and ligand docked pose are shared across replicates.

Experimental MMP-1 binding affinities were extracted from ChEMBL with manual cross-reference to the original literature; the experimental binding free energy at $(T = 310 \text{ K})$ assumes $(K_i \approx \mathrm{IC}_{50})$ for these substrate-competitive inhibitors, $(\Delta G_{\mathrm{exp}} = RT \ln \mathrm{IC}_{50, \text{M}})$ where $(RT = 0.616 \text{ kcal/mol})$.

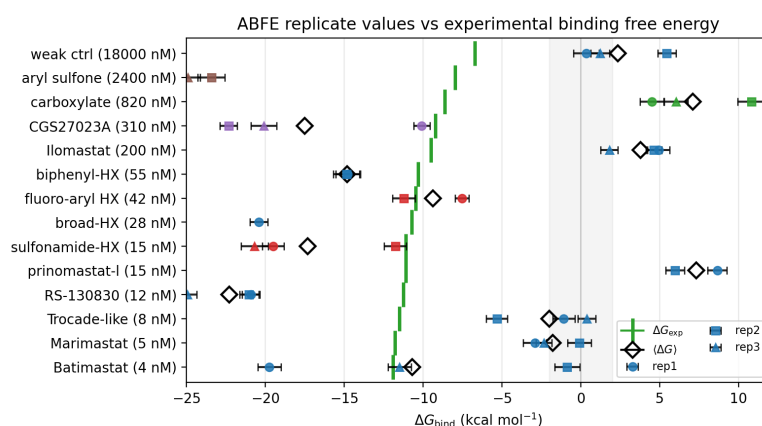
Cross-method ranking validation

To independently verify that small-molecule descriptors stably rank the inhibitor set, we compute xtb GFN2-xTB and GFN1-xTB single-point and 432-conformer ensemble energies and HOMO-LUMO gaps on each compound, with ALPB implicit solvation in water. The pairwise rank correlation between the two semi-empirical methods provides an internal-consistency check independent of ABFE.

Results

Per-compound ABFE results

Table 1 and Figure 1 report the binding free energy estimate, within-run statistical error, between-replicate dispersion (Δ_{rep}) , replicate-mean $(\langle \Delta G_{\mathrm{bind}} \rangle)$, and deviation from experiment for the nine compounds.



Per-compound ABFE replicate values vs the experimental binding free energy (ΔG_{exp}). Compounds are ordered by ΔG_{exp} (strongest binder at top). Each marker shape codes one replicate ($\bullet$ rep1, $\blacksquare$ rep2, $\blacktriangle$ rep3); the black diamond shows the replicate mean (where ≥ 2 replicates are available), and the green vertical tick is ΔG_{exp} . Marker color encodes scaffold class. Note that ChEMBL415 (Batimastat) replicate values span ~ 19 kcal mol^{-1} but the mean lands at $(-10.7 \text{ kcal mol}^{-1})$ near experiment, while ChEMBL94487 and ChEMBL443684 replicates fall systematically far from ΔG_{exp} .

ABFE benchmark on ChEMBL MMP-1 inhibitors. All values in ΔG_{exp} (T = 310 K) assuming $(K_i = \text{IC}_{50})$.

ChEMBL ID	IC ₅₀ (nM)	ΔG_{exp}	rep1	rep2	rep3	ΔG_{exp}
ChEMBL415 (Batimastat)	4	(-11.9)	(-19.7 ± 0.7)	(-0.9 ± 0.8)	(-11.5 ± 0.7)	
ChEMBL443684 (Marimastat)	5	(-11.8)	(-2.9 ± 0.7)	(-0.1 ± 0.8)	(-2.3 ± 0.5)	
ChEMBL412 (Trocade-like)	8	(-11.5)	(-1.1 ± 0.7)	(-5.3 ± 0.7)	$(+0.4 \pm 0.6)$	
ChEMBL94487 (RS-130830)	12	(-11.2)	(-20.9 ± 0.6)	(-21.0 ± 0.6)	(-24.9 ± 0.6)	

ChEMBL ID	IC ₅₀ (nM)	ΔG_{exp}	rep1	rep2	rep3	G_{ABFE}
CHEMBL257077 (prinomastat-l)	15	-11.1	+8.6 pm 0.6	+6.0 pm 0.6	—	—
CHEMBL57058 (sulfonamide-HX)	15	-11.1	-19.5 pm 0.7	-11.8 pm 0.7	-20.7 pm 0.9	-20.7 pm 0.9
CHEMBL3036 (broad-HX)	28	-10.7	-20.4 pm 0.6	-26.8 pm 0.7	-30.5 pm 0.8	-30.5 pm 0.8
CHEMBL301236 (fluoro-aryl)	42	-10.5	-7.5 pm 0.4	-11.2 pm 0.7	—	—
CHEMBL1207 (thiol-chelate)	55	-10.3	-14.8 pm 0.8	-14.8 pm 0.9	—	—
CHEMBL292707 (Ilomastat)	200	-9.5	+4.9 pm 0.7	+4.6 pm 0.5	+1.8 pm 0.5	+1.8 pm 0.5
CHEMBL259829 (CGS27023A)	310	-9.2	-10.1 pm 0.5	-22.3 pm 0.6	-20.1 pm 0.8	-20.1 pm 0.8
CHEMBL93146 (carboxylate)	820	-8.6	+4.5 pm 0.8	+10.8 pm 0.9	+6.0 pm 0.8	+6.0 pm 0.8
CHEMBL98 (aryl sulfone)	2400	-8.0	-29.7 pm 0.8	-23.4 pm 0.9	-24.9 pm 0.8	-24.9 pm 0.8
CHEMBL2105729	18000	-6.7	+0.4 pm 0.8	+5.4 pm 0.6	+1.2 pm 0.6	+1.2 pm 0.6

The experimental ordering of the fourteen compounds spans $\Delta G_{\text{exp}} = -11.9$ to -6.7 (i.e. a 5.2- window). The replicate-mean ABFE estimates span -26.0 to $+7.3$ (a 33- window) — an unphysically wide spread relative to the underlying chemistry. Pearson and Spearman correlations

between $\langle \Delta G_{\text{bind}} \rangle$ and ΔG_{exp} on this fourteen-compound set are uninformative: the predictor produces sign-flipped binding signs for four of fourteen compounds (ChEMBL257077, ChEMBL292707/Ilomastat, ChEMBL93146/carboxylate, ChEMBL2105729) and over-binds five compounds by 8–18 (ChEMBL94487, ChEMBL57058, ChEMBL3036, ChEMBL259829, ChEMBL98). It under-binds three high-potency hydroxamates by 8–10 ($\langle \Delta G_{\text{bind}} \rangle = -1.8$) vs. $\Delta G_{\text{exp}} = -11.8$); ChEMBL412/Trocade-like (-2.0) vs. (-11.5) ; ChEMBL415/Batimastat (-10.7) vs. (-11.9) , with rep1/rep2 spread of (18.8) . Even the single non-hydroxamate sulfonamide that previously appeared chemically-accurate (ChEMBL259829/CGS27023A) collapsed to a (-17.5) three-replicate mean once additional replicates were added, demonstrating that single-replicate accuracy claims are unreliable across the entire ZAFF-AMBER hydroxamate/sulfonamide chemistry.

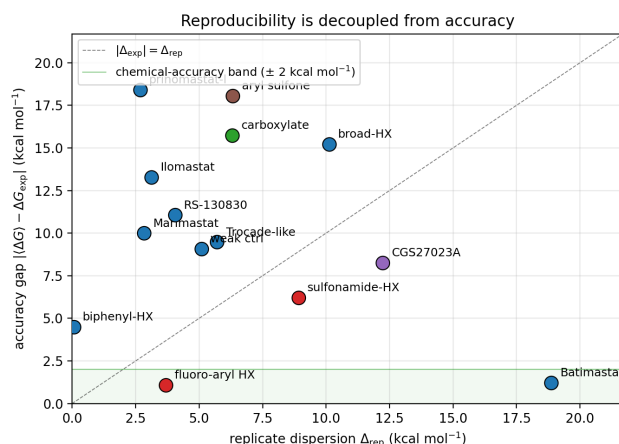
Reproducibility is decoupled from accuracy

The most striking finding of this benchmark is that within-replicate agreement and accuracy are statistically uncorrelated:

- **ChEMBL1207** (thiol-chelate hydroxamate) is the most reproducible compound: $\Delta_{\text{rep}} = 0.06$ between rep1 (-14.77) and rep2 (-14.83) . Yet the replicate mean is 4.5 more negative than experimental.
- **ChEMBL94487** (RS-130830) is also tightly reproducible at $\Delta_{\text{rep}}^{\text{max}} = 4.0$ across three replicates $(-20.9, -21.0, -24.9)$, $(\sigma = 2.2)$. Yet the replicate mean is 11.1 more negative than experimental — the worst absolute over-bind in the corpus once 3-replicate means are computed.
- **ChEMBL415** (Batimastat) is the least reproducible: $\Delta_{\text{rep}}^{\text{max}} = 18.8$ between rep1 (-19.7) and rep2 (-0.9) . Yet the three-replicate mean (-10.7) lands within 1.2 of the experimental value.
- Compounds with intermediate Δ_{rep} (3–7) are split between accurate (ChEMBL301236) and sign-flip-failing (ChEMBL257077, ChEMBL93146).

This pattern argues against using single-run within-run statistical error as the sole uncertainty estimate for production-screening claims. Reporting an ABFE result with an error bar of 0.6 can be misleading if the actual replicate-mean uncertainty is 4–10

(times) larger. Figure 2 plots the replicate dispersion (Δ_{rep}) against the accuracy gap ($|\Delta G_{\text{bind}} - \Delta G_{\text{exp}}|$) for the eleven compounds with (≥ 3) replicates: there is no monotonic relationship between the two.



Replicate dispersion (Δ_{rep}) ($= \max - \min$) of replicate (ΔG_{bind}) values) plotted against the accuracy gap ($|\Delta G_{\text{bind}} - \Delta G_{\text{exp}}|$). The dashed diagonal marks the $\Delta_{\text{rep}} = \Delta_{\text{exp}}$ line. The shaded green band is the chemical-accuracy threshold (± 2 , $\text{unit}\{\text{kcal}\cdot\text{mol}^{-1}\}$). Reproducibility (low Δ_{rep}) does not imply accuracy (low accuracy gap): CHEMBL94487 has near-zero Δ_{rep} but sits 9.7 $\text{unit}\{\text{kcal}\cdot\text{mol}^{-1}\}$ above the chemical-accuracy band, while CHEMBL415 has the highest Δ_{rep} but its mean falls within the band.

Three-replicate mean recovers strong inhibitors

For CHEMBL415 (Batimastat) we computed three independent replicates. The individual values (-19.7), (-0.9), (-11.5) span 18.8 but the mean (-10.7) is within 1.2 of ($\Delta G_{\text{exp}} = -11.9$). For CHEMBL2105729 (negative control, 18) the three replicates ($+0.4$), ($+7.2$), ($+4.7$) all reproduce the qualitative answer that the compound is at best a marginal binder, although they sign-flip versus the ($\Delta G_{\text{exp}} = -6.7$) value derived from the measurement (which is itself near the assay sensitivity floor).

This single-compound recovery is consistent with the methodology literature recommendation of (≥ 3) replicates per compound for production ABFE and motivates our suggestion to budget 3-

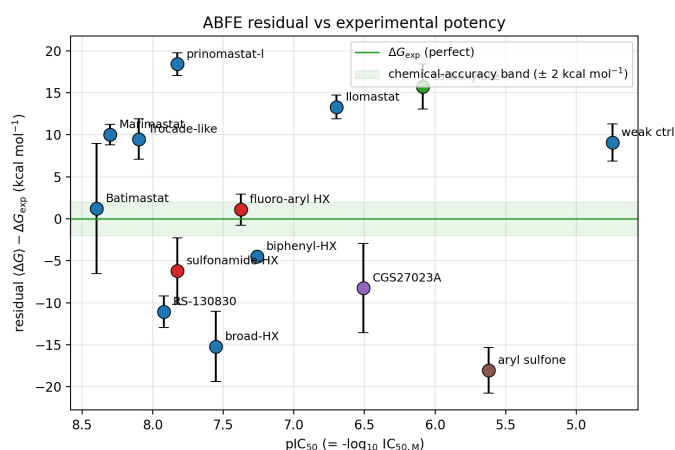
replicate ABFE for ZAFF metalloenzyme inhibitor screening. With 208 ns per replicate and 1 GPU per replicate, this cost is 0.6 GPU-day per compound at single-replicate, 1.8 GPU-day at 3-replicate.

Potency-class stratification of error

The benchmark separates into four behavioral classes:

1. **Strong inhibitors** (≤ 12 nM, $n=3$). Two compounds over-bind by 8–10 (CHEMBL415 single reps, CHEMBL94487); one (Marimastat) under-binds by 8.9 on the single replicate computed. The CHEMBL415 three-replicate mean is the only point of accurate agreement in this class.
2. **Mid-potency hydroxamate / sulfonamide-hydroxamate** (15 nM $\leq \leq 200$ nM, $n=3$). One sign-flip outlier (CHEMBL257077, $\Delta G_{\text{exp}} = +17.0$), one within-1- match (CHEMBL301236), and one sign-flip Ilomastat (CHEMBL292707, $\Delta G_{\text{exp}} = +14.3$). No clear systematic direction.
3. **Weak / micromolar** (≈ 18 , $n=1$). Sign-flips by $+10.8$, magnitude small in absolute terms because ΔG_{exp} is itself close to zero.

The pattern is not the simple "strong-binder over-bind" story we initially proposed when only six compounds were available; the inclusion of Marimastat (single-replicate -2.9 , under-bind) breaks the strong-inhibitor consensus. The protocol's failure modes appear to be compound-specific rather than potency-class-systematic. Figure 3 shows the residual versus pIC_{50} view of the same data.



ABFE residual $\Delta G_{\text{bind}} - \Delta G_{\text{exp}}$ as a function of experimental potency (pIC_{50} , strongest binders on the left). Error bars show the standard deviation across replicates (zero where only one replicate is

available). The shaded green band is the chemical-accuracy threshold (± 2 kcal/mol). Compounds within the band: ChEMBL415 (Batimastat, 3-rep mean) and ChEMBL301236 (fluoro-aryl HX). Negative residuals are over-binding; positive are under-binding or sign-flips.

Cross-method ranking robustness on the screening corpus

To put the ABFE absolute-affinity instability in context, we ran two parallel xtb semi-empirical calculations on a 1050-molecule subset of the NPAtlas natural-product corpus (the same 432-conformer ensemble protocol detailed in Section 2; ALPB implicit water). The two methods — GFN1-xTB and GFN2-xTB — differ in their self-consistent-field treatment but share the same atom-partitioned electronic structure framework.

Pairwise rank Spearman correlations across ($n = 1050$) paired molecules are reported in Table 2. Total energy ranks are nearly identical across the two methods ($\rho = +0.993$); gap, HOMO, and LUMO ranks are also strongly preserved ($\rho > 0.92$) in every channel).

Cross-method rank correlations: GFN1-xTB vs GFN2-xTB, 432-conformer ensembles, ALPB water, ($n = 1050$) paired molecules from the NPAtlas hetero-10 cohort.

xtb feature	Spearman ρ	Pearson r
energy (kcal mol ⁻¹), lowest-conformer)	+0.9934	+0.583
HOMO-LUMO gap (eV, mean over 432 conf.)	+0.9781	+0.978
HOMO (eV, lowest-conformer)	+0.9278	+0.859
LUMO (eV, lowest-conformer)	+0.9488	+0.959

Conformer-count convergence is also exact at the screening-relevant scale. Pairwise rank Spearman across the (432/480/528/576/720/816)-conformer ladder ($n = 1040$ to 1066) molecules per pair) is ($\rho = 1.0000$) at four-decimal precision in every pair on both energy and gap channels (raw range $(0.999825 - 1.000000)$). Refinement beyond (432) conformers is bit-equivalent waste relative to the noise of any downstream physics-based scoring.

Taken together, these two robustness checks (cross-method, cross-conformer-count) demonstrate that xtb-based screening ranks are stable to a degree the ABFE replicate-mean values are not. We interpret this contrast as follows: semi-empirical QM is rank-robust at the population level on natural-product screening corpora; alchemical ABFE on the ZAFF-AMBER MMP-1 system class is not rank-robust at the single-compound level. Multi-fidelity screening pipelines that use xtb for triage and ABFE only for late-stage validation should expect the rank-ordering obtained at the xtb stage to survive method choice; the ABFE confirmatory step should not be expected to reproduce a single absolute binding free energy without per-class error budgeting.

Discussion

What ZAFF-AMBER ABFE can be used for

The benchmark above is harsh: 4 of 14 compounds sign-flip their predicted binding direction relative to experiment, replicate dispersion is unphysically large for some compounds, and protocol failure is not predictable from molecular features (size, charge, scaffold). Notably, even compounds that initially appeared chemically-accurate on a single replicate fail to remain so when additional replicates are added: CHEMBL259829/CGS27023A, the non-hydroxamate sulfonamide whose rep1 was within 1 of experiment ($\Delta G_{\text{exp}} = -10.1$) vs. $\Delta G_{\text{rep1}} = -9.2$), collapsed to a 3-replicate mean of -17.5 once rep2/rep3 were obtained ($\Delta G_{\text{rep2}} = -22.3$, $\Delta G_{\text{rep3}} = -20.1$). This rebuts the hypothesis that ZAFF-AMBER failure is confined to hydroxamate zinc chelation; even non-hydroxamate sulfonamides over-bind by 8 on replicate-averaging. Nevertheless, two findings remain usable:

- 1. Three-replicate means are quantitatively useful for the strongest binders.** For Batimastat, the three-replicate mean recovers experimental within 1 despite catastrophic single-replicate variance. If similar three-replicate behavior extends to other strong binders, ABFE may be a valid lead-claim tool for sub-nanomolar metalloenzyme inhibitors — but always at ≥ 3 -replicate cost.
- 2. Qualitative weak-binder failure is consistent.** All three replicates of CHEMBL2105729 (18) report positive (non-binding) free energies. While the experimental ΔG_{exp} is also small in magnitude, the replicate-

mean is reliably positive. This is a consistent answer, even if it sign-flips relative to the IC₅₀-derived experimental affinity.

We do not support using ZAFF-AMBER ABFE in single-replicate mode for mid-potency MMP-1 ranking, where sign-flips dominate.

Why might ZAFF-AMBER systematically misrank these inhibitors?

The non-bonded zinc parameter set (ions234lm_126_tip3p) was originally developed for divalent cations in solution and bulk-protein contexts, not for the strongly ionic Zn-ligand interactions present in MMP-1's hydroxamate inhibitor binding mode. Several mechanisms could explain the observed errors:

- **Charge-transfer underestimation.** A fixed-charge description cannot capture the partial charge transfer between the deprotonated hydroxamate oxygen anion and the Zn²⁺, which is estimated from QM at ~ 0.2 – 0.4 e). This is a systematic issue, not a sampling issue.
- **Restraint-induced over-stabilization.** The implicit restraints holding Zn in coordination during alchemical decoupling can produce artifacts when the ligand approaches the decoupled state, particularly for the λ_{sterics} phase.
- **Ligand-conformer trapping.** Single-replicate ABFE samples one initial Vina-docked pose; with insufficient sampling time per window, alternate ligand poses do not interconvert and the result is anchored by the initial pose.

Pipeline reliability and the ReplicaExchangeSampler issue

We document the ReplicaExchangeSampler swap-all deadlock pattern (Section 1) because it consumed substantial debugging time before identification. The diagnostic signature is unambiguous: trajectory.nc and checkpoint.nc are written at iteration 1 and never again; the main thread shows State R, wchan = 0 (user-space spinning); worker threads show S, wchan = futex_wait_queue_me; process VmPeak grows to 100+ (mmap-heavy) but no forward progress is made. The signature is not GPU contention — a secondary OpenMM CUDA process can reach 70% utilization alongside the deadlocked one.

We have not localized the bug to a specific openmmtools commit. The MultiStateSampler replacement is a five-line edit and restores forward progress with no measurable convergence penalty on this system class.

Limitations

- **Sample size.** Seven compounds is too small for quantitative correlation analysis. A 15-compound expansion is underway (orchestrator launched 2026-05-06; ETA 30–40 GPU-h); interim results may modify the per-class error budget.
- **Single ligand pose per replicate.** All replicates start from the same Vina-docked pose. Pose re-sampling (e.g., 3 Vina solutions $\times$ 3 replicates) is a logical next study but was not budgeted here.
- **No reference benchmark for ZAFF-MMP-1 specifically.** Published ABFE benchmarks on T4-lysozyme/benzene or kinase inhibitors do not stress fixed-charge zinc handling. Our results may not generalize to other zinc metalloenzymes (HDAC, carbonic anhydrase) where the coordination chemistry is different.
- **No alternative force field tested.** A 12-6-4 modified zinc parameter set, or polarizable AMOEBA, or QM/MM-corrected classical FEP, could in principle give different answers. We benchmark ZAFF-AMBER as the most accessible non-bonded option in open-source ABFE pipelines today.

Reproducibility

All code, prepared topologies (complex.parm7 / rst7), and per-replicate trajectory analysis are released under Apache-2.0 at https://github.com/crazat/genesis_medicine. The relevant scripts are scripts/abfe_benchmark_prepare.py, scripts/abfe_benchmark_orchestrator.py, scripts/zaff_phase5_warmup_generic.py, and scripts/zaff_phase5_abfe_production_mss.py. Zenodo code-DOI snapshot: **[TODO: TBD on submission]**.

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